

Amenability of a twisted ℓ^1 -crossed product forces amenability of the acting group

Solution packet for arXiv:1705.02623

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Abstract

De Jeu, El Harti and Pinto ask whether amenability of $\ell^1(G, A; \alpha)$, for a discrete group G and a nonzero C^* -algebra A , forces G to be amenable. We prove that it does. In fact, the C^* -structure is unnecessary: the conclusion holds for every nonzero Banach algebra acted on isometrically by G . The proof turns a multiplier-central approximate diagonal into a Reiter P_1 net by retaining its (p, p^{-1}) -coefficients.

1 The source problem and the answer

For an action $\alpha : G \rightarrow \text{Aut}(A)$ of a discrete group by isometric automorphisms, write

$$\mathcal{B} = \ell^1(G, A; \alpha), \quad (f * h)(r) = \sum_{s \in G} f(s) \alpha_s(h(s^{-1}r)).$$

Section 5 of [1] asks whether amenability of $\mathcal{B}$, for nontrivial A , implies amenability of G . The exact statement appears on source PDF page 16:

The proofs of the first and third statements rely heavily on some non-trivial results for C^* -algebras. It is also possible to deduce the second statement in this fashion. As before, if $\ell^1(G)$ is amenable, then so is $C_r^*(G)$. Since G is discrete, [14, Theorem 4.2] then shows that G is amenable. Compared to this, Ringrose's argument for the amenability of G is definitely more direct, and it would be desirable to have a more direct argument for the first and third statements as well.

We do not know if, for non-trivial A , the amenability of $\ell^1(G, A; \alpha)$ implies the amenability of G . An attempt to derive such results via the enveloping C^* -algebra (as above) does not only seem overly complicated, but

Theorem 1 (Full answer, in stronger form). *Let $A \neq \{0\}$ be a Banach algebra and let the discrete group G act on A by isometric algebra automorphisms. If $\ell^1(G, A; \alpha)$ is an amenable Banach algebra, then G is an amenable group.*

This applies in particular to every nontrivial C^* -dynamical system in the question. No unit, character, invariant state, positivity, separability, or countability assumption is needed.

2 Proof intuition

An approximate diagonal for $\mathcal{B}$ is a family of two-variable coefficient tensors. The group itself acts on $\mathcal{B}$ by canonical left and right multipliers, even if A has no unit. Amenability lets us choose the diagonal asymptotically central under those multipliers.

For a diagonal tensor D_i , keep only

$$z_i(p) = D_i(p, p^{-1}) \in A \hat{\otimes} A.$$

Centrality under the multiplier indexed by g says, after looking at the output coordinate (gp, p^{-1}) , that $z_i(gp)$ is close in summed norm to $(\alpha_g \otimes \text{id})z_i(p)$. Since $\alpha_g \otimes \text{id}$ is an isometry, the nonnegative scalar masses $\|z_i(p)\|$ are almost translation invariant.

They cannot all disappear: the identity coefficient of the product $\pi(D_i)$ acts approximately as the identity on any fixed nonzero element of A , and its norm is bounded above by $\sum_p \|z_i(p)\|$. Normalising these masses therefore gives a Reiter P_1 net on G .

3 A multiplier-central diagonal lemma

For a Banach algebra C , let $M(C)$ denote its algebra of bounded double centralizers. If $T = (L_T, R_T) \in M(C)$, its actions on $X = C \hat{\otimes} C$ are

$$T \cdot (a \otimes b) = L_T(a) \otimes b, \quad (a \otimes b) \cdot T = a \otimes R_T(b).$$

Lemma 2 (Multiplier-central bounded approximate diagonal). *If C is an amenable Banach algebra and $\mathcal{T} \subset M(C)$, there is a bounded net $(D_i) \subset C \hat{\otimes} C$ such that*

$$\begin{aligned} \|T \cdot D_i - D_i \cdot T\| &\longrightarrow 0 & (T \in \mathcal{T}), \\ \|\pi(D_i)c - c\| &\longrightarrow 0 & (c \in C). \end{aligned}$$

Proof. Amenability gives a virtual diagonal $M \in (C \hat{\otimes} C)^{**}$: for $c \in C$,

$$c \cdot M = M \cdot c, \quad \pi^{**}(M)c = c.$$

It also gives a bounded two-sided approximate identity (e_λ) . Fix $T = (L_T, R_T) \in M(C)$. After taking a subnet, let $x_T \in C^{**}$ be a weak-star cluster point of $L_T(e_\lambda)$. The double-centralizer identities give, for $c \in C$,

$$x_T c = L_T(c), \quad c x_T = R_T(c),$$

because $L_T(e_\lambda)c = L_T(e_\lambda c) \rightarrow L_T(c)$ and $c L_T(e_\lambda) = R_T(c)e_\lambda \rightarrow R_T(c)$.

Extend the left and right module actions canonically to the biduals, taking the extension weak-star continuous in the multiplier variable. Passing from centrality under $L_T(e_\lambda) \in C$ to its weak-star cluster point shows

$$T \cdot M = x_T \cdot M = M \cdot x_T = M \cdot T. \tag{3.1}$$

This can also be checked directly after pairing with an arbitrary element of $(C \hat{\otimes} C)^*$; both sides are bounded linear functionals of $x_T \in C^{**}$.

It remains to return from the bidual to actual tensors. Given finite sets $F \subset \mathcal{T}$, $K \subset C$, consider the bounded affine map

$$\Phi(D) = ((T \cdot D - D \cdot T)_{T \in F}, (\pi(D)c - c)_{c \in K})$$

into the corresponding finite ℓ^∞ -sum of copies of $C\widehat{\otimes}C$ and C . By (3.1) and the virtual-diagonal identity, $\Phi^{**}(M) = 0$. Goldstine's theorem puts zero in the weak closure of the image under Φ of a fixed bounded ball in $C\widehat{\otimes}C$. That image is convex, so Hahn–Banach says that zero is also in its norm closure. Directing over (F, K, ε) gives the required bounded net. $\square$

4 Proof of the theorem

Put $\mathcal{B} = \ell^1(G, A; \alpha)$. For each $g \in G$, the formal element supported at g with multiplier-algebra coefficient 1 defines an isometric double centralizer $u_g \in M(\mathcal{B})$:

$$(u_g f)(r) = \alpha_g(f(g^{-1}r)), \quad (f u_g)(r) = f(r g^{-1}). \quad (4.1)$$

Apply Lemma 2 with $\mathcal{T} = \{u_g : g \in G\}$, obtaining a bounded net (D_i) .

There is a canonical isometric identification

$$\mathcal{B}\widehat{\otimes}\mathcal{B} \cong \ell^1(G \times G, A\widehat{\otimes}A).$$

Write $D_i(p, q)$ for the corresponding coefficient and set

$$z_i(p) = D_i(p, p^{-1}), \quad S_i = \sum_{p \in G} \|z_i(p)\|.$$

At the output coordinate (gp, p^{-1}) , formulas (4.1) give

$$(u_g \cdot D_i)(gp, p^{-1}) = (\alpha_g \otimes \text{id})z_i(p), \quad (D_i \cdot u_g)(gp, p^{-1}) = z_i(gp).$$

Consequently,

$$\sum_{p \in G} \|(\alpha_g \otimes \text{id})z_i(p) - z_i(gp)\| \leq \|u_g \cdot D_i - D_i \cdot u_g\| \longrightarrow 0. \quad (4.2)$$

We next show that S_i stays away from zero. Choose $a \in A$ with $\|a\| = 1$, and let $c_i \in A$ be the identity coefficient of $\pi(D_i) \in \mathcal{B}$. Since $\pi(D_i)$ is a left approximate identity for $\mathcal{B}$, applying it to the element supported at the identity with coefficient a gives

$$c_i a \longrightarrow a. \quad (4.3)$$

For $p \in G$, let $\theta_p = m_A \circ (\text{id} \otimes \alpha_p) : A\widehat{\otimes}A \rightarrow A$. Each θ_p is contractive, and the twisted multiplication formula gives the absolutely convergent identity

$$c_i = \sum_{p \in G} \theta_p(z_i(p)). \quad (4.4)$$

Thus $S_i \geq \|c_i\| \geq \|c_i a\| \rightarrow 1$. Also $S_i \leq \|D_i\|$, so the masses are finite and uniformly bounded.

For all sufficiently large i , define a probability function on G by

$$\mu_i(p) = \frac{\|z_i(p)\|}{S_i}.$$

Because $\alpha_g \otimes \text{id}$ is an isometry, (4.2) yields

$$\begin{aligned} \|\lambda_g \mu_i - \mu_i\|_1 &= \frac{1}{S_i} \sum_{p \in G} \left| \|z_i(gp)\| - \|z_i(p)\| \right| \\ &\leq \frac{1}{S_i} \sum_{p \in G} \|z_i(gp) - (\alpha_g \otimes \text{id})z_i(p)\| \longrightarrow 0. \end{aligned}$$

Hence (μ_i) is a Reiter P_1 net. The Reiter criterion for discrete groups now implies that G is amenable. $\square$

5 Boundary checks and what the proof does not claim

- The nonzero hypothesis is necessary for the formulation: if $A = \{0\}$, then $\ell^1(G, A; \alpha) = \{0\}$ contains no information about G .
- Nonunital A causes no problem. The symbols u_g live in $M(\mathcal{B})$, and Lemma 2 is precisely what permits their use.
- No scalar quotient of $\mathcal{B}$ is constructed. Such a quotient would usually require an α -invariant character, which need not exist.
- Isometry of the action is used only to pass from (4.2) to scalar norm invariance. Every C^* -automorphism is isometric.
- The result is the converse in the group variable only. The source already proves that amenability of $\mathcal{B}$ forces amenability of A ; the present argument does not strengthen that conclusion to strong amenability.

6 Novelty and provenance audit

The proof above is self-contained apart from the standard virtual-diagonal and Reiter characterisations of amenability. Ross Stokke’s scalar group-algebra work [4] is conceptually related: it also extracts group invariance from approximate diagonals. It neither states nor proves the coefficient-valued crossed-product converse here.

A bounded search through 27 August 2026 covered the exact arXiv id and title, exact phrases from the source question, combinations of $\ell^1(G, A; \alpha)$, amenability, converse, and group amenability, author/citation searches, and related Fell-bundle and convolution-dominated operator literature. No later explicit solution of the question was located. The result is therefore presented as a plausible new full answer, not as a priority claim.

7 Human-review focus

The two high-value checks are:

1. verify Lemma 2, particularly the weak-star passage from algebra centrality to double-centralizer centrality;
2. recompute the two coefficients at (gp, p^{-1}) from (4.1).

Once these are confirmed, (4.3)–(4.4) prevent loss of mass and the final Reiter argument is immediate.

References

- [1] M. de Jeu, R. El Harti, and P. R. Pinto, *Amenable crossed product Banach algebras associated with a class of C^* -dynamical systems. II*, arXiv:1705.02623 (2017), Section 5.
- [2] B. E. Johnson, *Cohomology in Banach algebras*, Memoirs of the American Mathematical Society 127 (1972).
- [3] V. Runde, *Lectures on Amenability*, Lecture Notes in Mathematics 1774, Springer, 2002.
- [4] R. Stokke, *Approximate diagonals and Følner conditions for amenable group and semigroup algebras*, Studia Mathematica 164 (2004), 139–159.